

GATE XL

Life Sciences

TREND-BASED MOCK PAPER

— 03 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	10 subjects

Difficulty Distribution

Easy: 20 questions (31%)

Moderate: 31 questions (48%)

Difficult: 14 questions (22%)

GENERAL INSTRUCTIONS

GATE XL — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[Cell Biology]

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

Q2. 1M [no -ve]

[Evolution]

Natural selection proposed by:

Enter your answer here

Q3. 1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

Q4. 1M [no -ve]

[Biochemistry]

Enzyme specificity is:

Enter your answer here

Q5. 1M [-0.33]

[Evolution]

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

Q6. 1M [-0.33]

[Genetics]

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) Random

Q7. 1M [no -ve]

[Ecology]

Trophic level of producers:

Enter your answer here

Q8.

1M [no -ve]

[Cell Biology]

Powerhouse of cell:

*Enter your answer here***Q9.**

1M [no -ve]

[Ecology]

Trophic level of producers:

*Enter your answer here***Q10.**

1M [-0.33]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q11.

1M [no -ve]

[Zoology]

Heart of mammal is:

*Enter your answer here***Q12.**

1M [-0.33]

[Ecology]

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

Q13.

1M [-0.33]

[Genetics]

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) Random

Q14.

1M [no -ve]

[Botany]

Photosynthesis produces:

Enter your answer here

Q15.

1M [-0.33]

[Genetics]

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) Random

Q16.

1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

Q17.

1M [no -ve]

[Botany]

Photosynthesis produces:

*Enter your answer here***Q18.**

1M [-0.33]

[Evolution]

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

Q19.

1M [-0.33]

[Ecology]

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

Q20.

1M [-0.33]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂

Q21.

1M [no -ve]

[Cell Biology]

Powerhouse of cell:

*Enter your answer here***Q22.**

1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

Q23.

1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

Q24.

1M [-0.33]

[Microbiology]

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

Q25.

1M [-0.33]

[Engineering Mathematics]

Mean of normal distribution is:

- (A) Median
- (B) Mode
- (C) Both
- (D) None

Q26.

1M [no -ve]

[Botany]

Photosynthesis produces:

*Enter your answer here***Q27.**

1M [-0.33]

[General Aptitude]

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

Q28.

1M [-0.33]

[Genetics]

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) Random

Q29.

1M [-0.33]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q30.

1M [no -ve]

[Zoology]

Heart of mammal is:

*Enter your answer here***Q31.**

2M [no -ve]

[Evolution]

Natural selection proposed by:

*Enter your answer here***Q32.**

2M [no -ve]

[Botany]

Photosynthesis produces:

*Enter your answer here***Q33.**

2M [no -ve]

[Ecology]

Trophic level of producers:

*Enter your answer here***Q34.**

2M [-0.66]

[Evolution]

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

Q35.

2M [-0.66]

[Evolution]

Natural selection proposed by:

- (A) Lamarck
- (B) Darwin
- (C) Mendel
- (D) Wallace

Q36.

2M [-0.66]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q37.

2M [no -ve]

[Biochemistry]

Enzyme specificity is:

*Enter your answer here***Q38.**

2M [-0.66]

[Biochemistry]

Enzyme specificity is:

- (A) High
- (B) Low
- (C) Zero
- (D) Variable

Q39.

2M [-0.66]

[Microbiology]

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

Q40.

2M [-0.66]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂

Q41.

2M [no -ve]

[Microbiology]

Penicillin from:

*Enter your answer here***Q42.**

2M [-0.66]

[General Aptitude]

Choose synonym of 'abundant':

- (A) Scarce
- (B) Plentiful
- (C) Minimal
- (D) Rare

Q43.

2M [no -ve]

[Microbiology]

Penicillin from:

*Enter your answer here***Q44.**

2M [-0.66]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q45.

2M [-0.66]

[Genetics]

DNA replication is:

- (A) Conservative
- (B) Semi-conservative
- (C) Dispersive
- (D) Random

Q46.

2M [-0.66]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q47.

2M [-0.66]

[Microbiology]

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

Q48.

2M [-0.66]

[Ecology]

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

Q49.

2M [no -ve]

[Cell Biology]

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

Q50.

2M [-0.66]

[Biochemistry]

Enzyme specificity is:

- (A) High
- (B) Low
- (C) Zero
- (D) Variable

Q51.

2M [no -ve]

[Ecology]

Trophic level of producers:

*Enter your answer here***Q52.**

2M [no -ve]

[Zoology]

Heart of mammal is:

*Enter your answer here***Q53.**

2M [-0.66]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂

Q54.

2M [no -ve]

[Cell Biology]

Powerhouse of cell:

Enter your answer here

Q55.

2M [no -ve]

[Evolution]

Natural selection proposed by:

*Enter your answer here***Q56.**

2M [-0.66]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂

Q57.

2M [no -ve]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂

Q58.

2M [no -ve]

[Microbiology]

Penicillin from:

*Enter your answer here***Q59.**

2M [-0.66]

[Cell Biology]

Powerhouse of cell:

- (A) Nucleus
- (B) Mitochondria
- (C) Ribosome
- (D) ER

Q60.

2M [-0.66]

[Ecology]

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

Q61.

2M [no -ve]

[Zoology]

Heart of mammal is:

- (A) 2-chambered
- (B) 3-chambered
- (C) 4-chambered
- (D) 5-chambered

Q62.

2M [-0.66]

[Microbiology]

Penicillin from:

- (A) Fungus
- (B) Bacteria
- (C) Virus
- (D) Algae

Q63.

2M [no -ve]

[Evolution]

Natural selection proposed by:

*Enter your answer here***Q64.**

2M [-0.66]

[Ecology]

Trophic level of producers:

- (A) 1st
- (B) 2nd
- (C) 3rd
- (D) 4th

Q65.

2M [-0.66]

[Botany]

Photosynthesis produces:

- (A) O₂
- (B) CO₂
- (C) N₂
- (D) H₂